

High-Level Design

The connected system behind small missions and big futures.



PURPOSE

A safe, adaptive digital coach for children ages 8–10. The app turns screen time into short activities that grow learning, creativity, and kindness.

The three product paths



LEARN

Logic, memory,
numeracy, focus



CREATE

Imagination,
storytelling, ideas



CONNECT

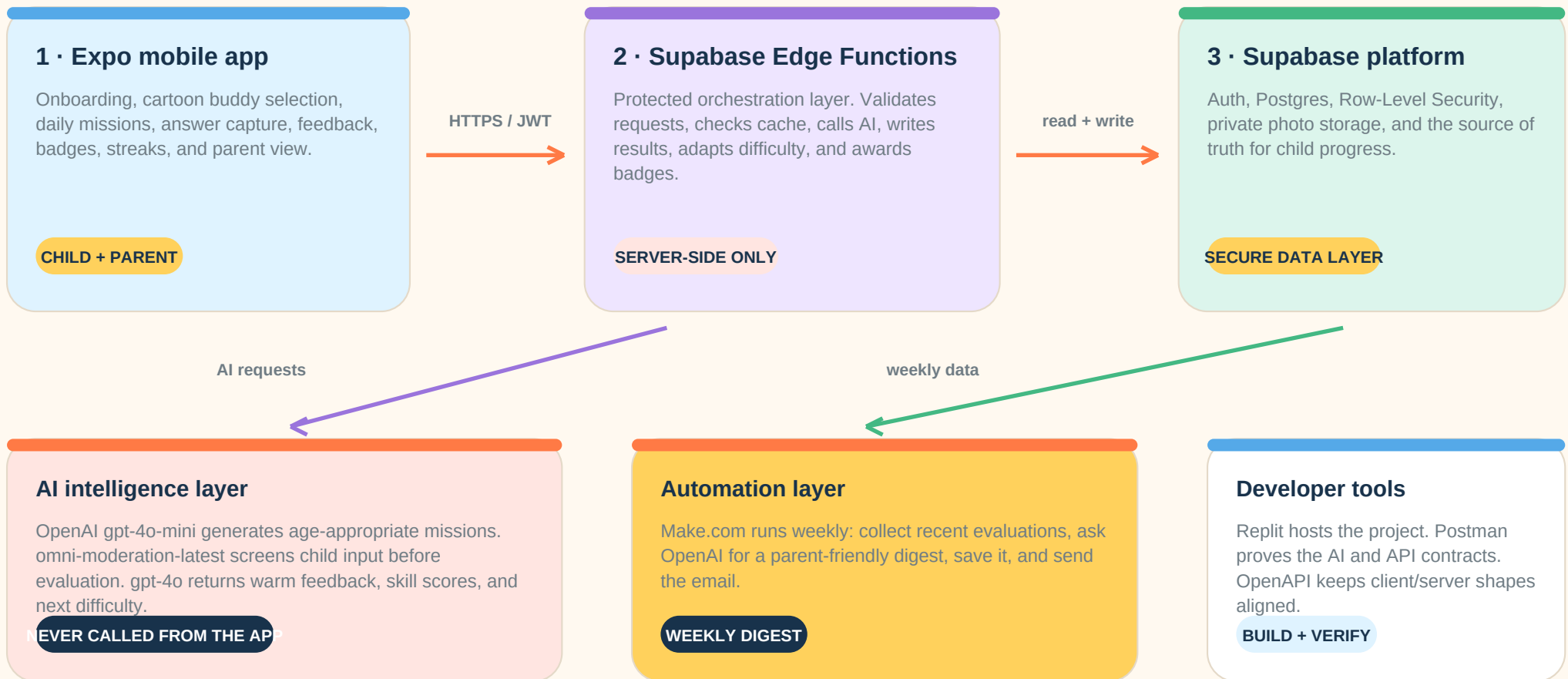
Kindness,
empathy,
teamwork

PRIMARY SUCCESS METRIC

A child completes a 15-minute mission, receives warm feedback within 8 seconds, and the parent view reflects a new insight.

One safe loop from child input to parent insight

The mobile app stays simple. Supabase owns data and server-side logic. OpenAI is accessed only from protected functions.



Key rule: the OpenAI key, service-role key, and child data operations stay behind server-side controls.

The modules needed to develop the full product

Each module owns one responsibility. This keeps the child experience fast, the data private, and the AI replaceable.

Mobile experience

- **Onboarding**
Name, age, interests, and cartoon companion.
Collect only minimum child data.
- **Mission home**
Three daily pillars, progress, streak, and stars.
- **Mission runner**
Instructions, text or photo response, submit state, and retry state.
- **Feedback + badges**
Warm child feedback, badge animation, difficulty update, and next step.
- **Parent view**
Skill paths, insights, badge shelf, and weekly note.

Supabase backend

- **Auth**
Parent identity and session token.
- **Postgres**
Parents, children, profiles, missions, responses, evaluations, badges, digests.
- **RLS**
Every child-scoped row is accessible only to the owning parent.
- **Storage**
Private bucket for optional child photos; signed URLs only.
- **Edge Functions**
generate-mission, submit-response, get-progress.

AI + operations

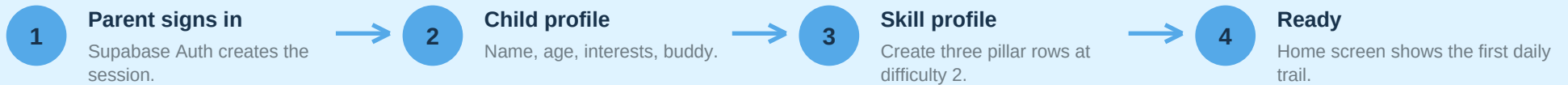
- **Mission generator**
gpt-4o-mini; adapts to age, interests, pillar, and difficulty.
- **Safety gate**
omni-moderation-latest runs before evaluation.
- **Evaluator**
gpt-4o; returns praise, insight, skills, effort, and next difficulty.
- **Weekly digest**
Make.com assembles and sends a plain-language parent update.
- **Testing + delivery**
Postman, Replit, Expo Go, and OpenAPI/codegen.

Recommended ownership Expo owns interaction and presentation. Supabase owns identity, data, and protected orchestration. OpenAI owns language intelligence. Make.com owns scheduled communication.

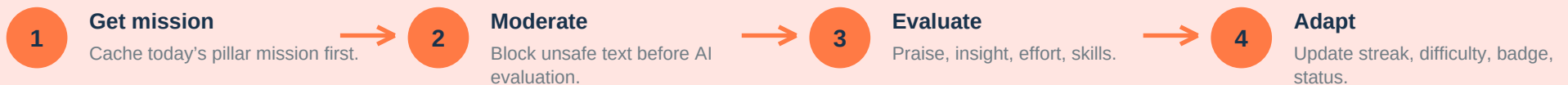
Three journeys that prove the product works

The full app is valuable because each flow ends in a visible outcome: a child action, a safer evaluation, or parent understanding.

A · FIRST-TIME SETUP



B · DAILY MISSION LOOP



C · WEEKLY PARENT DIGEST



Build order, guardrails, and what stays out of MVP

Start with one complete vertical slice. Then add polish, parent context, and automation without widening the product too early.

Recommended build order

- 01 Foundation** Supabase project, Auth, schema, RLS, private storage.
- 02 AI proof** Test generation, moderation, and evaluator JSON in Postman.
- 03 Core functions** Generate, submit, and progress endpoints.
- 04 Mobile vertical slice** Onboarding → mission → answer → feedback.
- 05 Parent context** Insights, badges, streaks, and weekly digest.

Non-negotiable guardrails

- No OpenAI call from the mobile client.
- RLS enabled before real child data.
- Private photo bucket; signed URLs only.
- Moderation before evaluation.
- Cache one mission per pillar per day.
- Collect first name and age only.

Deliberately out of scope for the first release

Audio-first missions

Start with text and optional photo upload; add voice after the core loop is reliable.

Payments and schools

No subscriptions, school accounts, or leaderboards in the MVP.

Multiple children

Keep one child profile per parent until the parent experience is proven.

Multi-language content

Keep one language and one age band so tone and safety are easier to validate.